Highest order derivative is $\frac{d^2y}{dx^2}$

$\therefore$ order = 2

Power of the highest order derivative $\frac{d^2y}{dx^2}$ is 1.

$\therefore$ Degree =1

(ii) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 3y = 0$

Highest order derivative is $\frac{d^2y}{dx^2}$

$\therefore$ order =2

Power of the highest order derivative $\frac{d^2y}{dx^2}$ is 1.

$\therefore$ Degree = 1

(iii) $\frac{d^3y}{dx^3} - 3\left(\frac{dy}{dx}\right)^6 + 2y = x^2$

$\therefore$ order =3, Degree =1

(iv) $\left[1+\frac{d^2y}{dx^2}\right]^{\frac{3}{2}} = a\frac{d^2y}{dx^2}$

Here we eliminate the radical sign.

Squaring both sides, we get

$$\left[1+\frac{d^2y}{dx^2}\right]^3 = a^2\left(\frac{d^2y}{dx^2}\right)^2$$

$\therefore$ Order=2, degree =3

(v) $y' + (y'')^2 = (x + y'')^2$

$$y' + (y'')^2 = x^2 + 2xy'' + (y'')^2$$

$$y' = x^2 + 2xy'' \Rightarrow \frac{dy}{dx} = x^2 + 2x\frac{d^2y}{dx^2}$$

$\therefore$ Order=2, degree=1

(vi) $\frac{d^3y}{dx^3} - \left(\frac{dy}{dx}\right)^{\frac{1}{2}} = 0$

Here we eliminate the radical sign.

For this write the equation as

$$\frac{d^3y}{dx^3} = \left(\frac{dy}{dx}\right)^{\frac{1}{2}}$$

Squaring both sides, we get

$$\left(\frac{d^3y}{dx^3}\right)^2 = \frac{dy}{dx}$$

$\therefore$ Order=3, degree=2

(vii) $y = 2\left(\frac{dy}{dx}\right)^2 + 4x\frac{dx}{dy}$

$$y = 2\left(\frac{dy}{dx}\right)^2 + 4x\frac{1}{\left(\frac{dy}{dx}\right)}$$

$$y\frac{dy}{dx} = 2\left(\frac{dy}{dx}\right)^3 + 4x$$

$\therefore$ order=1, degree=3



Order and degree (if defined) of a differential equation are always positive integers.

Family of Curves

Sometimes a family of curves can be represented by a single equation with one or more arbitrary constants. By assigning different values for constants, we get a family of curves. The arbitrary constants are called the parameters of the family.

For example,

(i) $y^2 = 4ax$ represents the equation of a family of parabolas having the origin as vertex where 'a' is the parameter.

(ii) $x^2 + y^2 = a^2$ represents the equation of family of circles having the origin as centre, where 'a' is the parameter.

(iii) $y = mx + c$ represents the equation of a family of straight lines in a plane, where m and c are parameters.

XII Std - Business Maths & Stat EM Chapter 4.indd 82 2/28/2020 6:01:10 PM